MIL-B-37811A
29 September 1978
SUPERSEDING
MIL-B-37811(DLA-DM)
18 November 1977

MILITARY SPECIFICATION

BURS, DENTURE TRIMMING, STRAIGHT HANDPIECE, TUNGSTEN CARBIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tungsten carbide denture trimming burs used for trimming, carving, and finishing dentures in prosthetic laboratory procedures.

1.2 Classification. The denture trimming burs covered by this specification shall be of the following shapes, as specified (see 6.1):

Shape 1. - Flame.
Shape 2. - Pear.
Shape 3. - Round.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

MIL-P-116 - Preservation-Packaging, Methods Of.

STANDARDS

FEDERAL

FED-STD-66 - Steel, Chemical Composition and Hardenability.
FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24393 - Bur, Denture Trimming, Straight Handpiece, Tungsten Carbide.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Bur head. The bur head shall be fabricated of cemented tungsten carbide with a cobalt binder. Titanium carbide or tantalum carbide, or both, may be present to achieve the required properties.

3.1.2 Shank. The shank shall be fabricated of AISI 4130 or 4340 steel in accordance with FED-STD-66 for composition only.

3.2 Hardness. The bur head shall have a Rockwell hardness of A92 ± 2. The shank shall have a Rockwell hardness of C43 to C48, when tested as specified in 4.5.2.

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3.3 Design and construction. The bur shape shall be as shown in Defense Personnel Support Center (DPSC) Drawing No. 24393. The burs shall be straight handpiece burs for trimming, carving, and finishing acrylic, chrome alloy, and gold appliances. Burs shall be furnished in shapes as specified. The burs shall be of two-piece construction, and shall consist of a steel shank and carbide head. The shank of the bur shall be straight. The blades of the bur head shall be shaped and sharpened to cut at the tip as well as the sides of the bur head, and to provide automatic chip clearance. The joint of the bur shall be securely and permanently welded or brazed.

3.4 Runout. The total indicated runout of the burs shall not exceed 0.003 inch, when tested as specified in 4.5.3.

3.5 Fatigue in cutting. The time required to cut the eighteenth one-half inch shall be not more than 2.25 times the time that is required to cut the second one-half inch, when tested as specified in 4.5.4.1 and 4.5.4.2. The burs shall show no breaking, bending, or loosening of the head as a result of these tests.

3.6 Finish. The bur heads shall have a finely honed finish with sharpened blades. The joint and shank of the burs shall be smoothly finished.

3.7 Identification marking. Marking of the burs with the name or registered trademark of the manufacturer is optional.

3.8 Drawings. The drawings forming a part of this specification are not production drawings. Drawings established dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit, and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.9 Workmanship. Workmanship shall be first class throughout. The burs shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary, to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II, and III (see 4.3.1, 4.3.2, and 4.4).

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. The unit of product for examination purposes shall be one bur.

TABLE I. Sampling for Examination .

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor	II	4.0

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one bur.

TABLE II. Sampling for Test.

Characteristic	Require-ment	Test procedure	Inspection level
Chemical composition	3.1	4.5.1	(see 4.3.3)
Hardness	3.2	4.5.2	S-2
Runout	3.4	4.5.3	S-3
Fatigue in cutting (side of head)	3.5	4.5.4.1	S-2
Fatigue in cutting (tip of head)	3.5	4.5.4.2	S-2

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4.3.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

4.3.4 Sample unit usage. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

4.4 Examination. The burs shall be examined for the possible defects listed in table III. The examination shall not be limited to those defects shown in the table. Unless otherwise indicated in the table, the defect listed shall apply to all shapes of denture trimming burs covered by this specification.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Bur head shape not as specified.
102	Bur shank shape not as specified.
103	Head diameter does not comply with requirements at maximum cross-section.
104	Diameter of shank exceeds specification limits.
105	Length of head does not comply with requirements for applicable bur.
106	Burs not free from defects such as chipped blades, cracks, or pits.
107	Number of blades not as specified.
Minor	
201	Finish on bur not as specified.
202	Overall length of bur not within specification limits.
203	Shank dimensions, other than those specified under major defects, not within specification limits.
204	Welded or brazed joint of bur does not give the appearance of an integral head unit.

4.4.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.4.2 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10X) magnification.

4.4.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples, selected as specified in 4.3.3, shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

4.5.2 Hardness. Hardness shall be determined in accordance with Method 243 (Rockwell) or Method 244 (Diamond Pyramid) of FED-STD-151.

4.5.3 Runout. The bur shall be placed and held in a "V" block or precision chuck and rotated slowly. A dial indicator, sensitive to 0.0002 inch and equipped with a knife-edge foot, a shadow-graph, or other suitable equipment of equal accuracy shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.5.4 Fatigue in cutting.

4.5.4.1 Side of head. Tests shall be made on a milling machine or similar device. The feed shall be that produced by a 2-pound weight operating a cord attached to a carriage mounted on ball bearings. The bur shall rotate at 12,000 revolutions per minute. The material to be cut shall be 0.020 inch thick rolled brass with a Brinell hardness of 74-81 (Rockwell 30-T, 42-49). The edge of the brass shall be centered on the length of the head of the bur. The bur shall be perpendicular to the brass sheet and the cut shall be made the length of the brass. A recording device operated by a synchronous motor shall be used to time the cut. The brass sheet shall be cut for a total of 9 inches and the time taken to cut the eighteenth one-half inch and the second one-half inch shall be compared.

4.5.4.2 Tip of head. The test shall be conducted as specified in 4.5.4.1, except that the brass test piece shall be applied against the point of the bur.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One bur of the shape specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Burs shall be cleaned in accordance with MIL-P-116, method C-1.

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5.1.2.2 Drying. Burs shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Each bur shall have the shank coated with a suitable preservative oil and the head protected with a suitable peel-type protective coating.

5.1.2.4 Unit packaging. Each bur shall be packaged in a plastic vial or box having a friction closure. The burs shall be packaged in such a manner as to prevent damage to the cutting head. Each vial or box shall be further packaged in accordance with MIL-P-116, method IC-1.

5.1.2.5 Intermediate package. Six unit packages shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Burs shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Ninety-six intermediate packages (576 units), containing burs of one shape only, shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to the applicable paragraphs of the appendix thereto. A case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Ninety-six intermediate packages (576 units), containing burs of one shape only, shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged burs shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Shipping container. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Shape required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-01-027-0344	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide: Flame shape.
6520-01-027-0345	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide: Pear shape.
6520-01-027-0346	BUR, DENTURE TRIMMING, Straight Handpiece, Tungsten Carbide: Round shape.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD-MB

Preparing activity:
DLA-DM

Project No. 6520-1479

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-37811A BURS, DENTURE TRIMMING, STRAIGHT HANDPIECE, TUNGSTEN CARBIDE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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